UTM UNIVERSITI TEKNOLOGI MALAYSIA RESEARCH UNIVERSITY	**UNIVERSITI TEKNOLOGI MALAYSIA**

Probability & Statistical Data Analysis (SECI1143)

PROJECT 2

Group : 06

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1.0 Introduction

Sleep plays a vital role in maintaining both physical and mental health, yet poor sleep quality and irregular sleep patterns are common issues in today's fast-paced lifestyles. Inadequate or disrupted sleep can contribute to a range of health problems, such as weakened immunity, heart disease, weight gain, anxiety, and depression. On the other hand, healthy sleep habits are linked to improved mood, better memory, and a reduced risk of chronic illnesses. Given its broad impact, understanding the connection between sleep and overall well-being is essential for encouraging healthier living.

To support this investigation, we are using the Health and Sleep Relation dataset from Kaggle, prepared by research scientist Orvile. The dataset includes 374 samples and provides detailed information on individuals' sleep habits, health conditions, and lifestyle factors. It is designed to support research on sleep hygiene, chronic illnesses, mental health, and lifestyle influences on sleep. The dataset is under the Health Promotion and Health Maintenance category funded by the University of Oxford (OXFDU1245).

In this project, we will apply a range of statistical analysis techniques, including one-sample hypothesis testing, correlation and regression analysis, and also Chi-Square Test of Independence. These methods will help us examine whether certain variables are related and identify patterns that reveal how different aspects of sleep—such as quality and duration—are associated with health outcomes.

The main objectives of this project are:

- To apply and perform statistical tests on a secondary dataset related to sleep and health.
- To examine the dependency relationships among selected variables in the dataset.

By achieving these objectives, we aim to uncover meaningful insights into how sleep patterns affect physical and mental health, contributing to a better understanding of sleep's role in promoting overall well-being.

2.0 Data Set

2.1 Description of data

Population: People around the world

Sample: 374 people

Table 2.1.1: Type of variable and Measurement Level

Variables	Type of Variable	Measurement Level
Person ID (person from 1 until 374)	Qualitative	Nominal
Gender (gender of the person)	Qualitative	Nominal
Age (age of the person)	Quantitative	Ratio
Occupation (occupation of the person)	Qualitative	Nominal
Sleep Duration (sleep duration in hours of the person)	Quantitative	Ratio
Quality of Sleep (sleep quality reported by the person)	Qualitative	Ordinal
Physical Activity Level (physical activity level reported by the person)	Qualitative	Ordinal
Stress Level (stress level reported by the person)	Qualitative	Ordinal
BMI Category (overweight, normal weight or other)	Qualitative	Ordinal
Blood Pressure (blood pressure of the person)	Quantitative	Ratio
Heart Rate (heart rate of the	Quantitative	Ratio

person)		
Daily Steps (steps per day of the person)	Quantitative	Ratio
Sleep Disorder (whether the person having any type of sleep disorder)	Qualitative	Nominal

2.2 Statistical Test Analysis

Table 2.2.1: Selected Variables with objectives, test analysis and expected outcome

Selected Variables	**Objectives**	**Test Analysis and Expected Outcome**
Sleep Duration	Test if average sleep duration is less than the recommended 8 hours per day	**Analysis:** 1-sample hypothesis test (Test on mean, variance unknown) **Possible outcome:** By conducting a one-sample t-test at a 0.05 significance level, we can determine whether the mean sleep duration of the sample is significantly less than the recommended 8 hours per day. If the test statistic is less than the critical value, or if the p-value is below 0.05, we reject the null hypothesis and conclude that the average sleep duration is significantly less than 8 hours.
Sleep Duration, Stress Level	To find out if there is a relationship between how long a person sleeps and their	**Analysis:** Correlation analysis

	stress level.	**Expected Outcome:** By calculating the correlation coefficient between sleep duration and stress level, we can determine the strength and direction of their relationship. If the correlation coefficient is negative and statistically significant, it would suggest that people who sleep longer tend to have lower stress levels. Conversely, a weak or non-significant correlation would indicate little to no relationship between these two variables.
Physical Activity Level, Quality of Sleep	To determine if an individual's reported physical activity level is a significant predictor of their perceived quality of sleep	**Analysis:** Regression analysis **Expected Outcome:** By conducting a regression analysis between physical activity level and perceived quality of sleep, we can determine whether physical activity is a significant predictor of sleep quality. If the regression coefficient for physical activity level is positive and statistically significant, it would suggest that individuals who engage in higher levels of physical activity tend to report better quality of sleep. Conversely, a non-significant or near-zero

		regression coefficient would indicate that physical activity level is not a meaningful predictor of sleep quality in this dataset.
Gender, Type of Sleep Disorder	To determine whether gender and type of sleep disorder are independent of each other.	**Analysis:** Chi-Square Test of Independence **Expected Outcome:** By calculating the Chi-Square (χ^2) statistic between Gender and Type of Sleep Disorder, we can assess whether there is a significant association between these two variables. If the Chi-Square value is large and exceeds the critical value, it indicates that the observed frequencies differ substantially from the expected frequencies. This suggests a statistically significant relationship, meaning that Gender and Type of Sleep Disorder are not independent and that the distribution of sleep disorders varies between males and females.

3.0 Data Analysis

3.1 One-Sample Hypothesis Test

In this analysis, we will examine the variable Sleep Duration to determine whether there is sufficient statistical evidence to support the claim that the mean sleep duration of the population is less than the recommended 8 hours per day, at a 95% confidence level. The dataset consists of a sample of 374 individuals. A one-sample t-test will be conducted to test the population mean, as the population standard deviation is unknown.

Hypothesis Statement:

Null Hypothesis (H_0): $\mu = 8$
(The population mean sleep duration is equal to 8 hours per day)

Alternative Hypothesis (H_1): $\mu < 8$ (claim)
(The population mean sleep duration is less than 8 hours per day)

Test Statistic:

Given a 95% confidence level, $\boldsymbol{\alpha}$=0.05. The test statistic, t_0 can be calculated using the formula:

$$t_0 = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

By using RStudio, the mean ($\bar{x}$) and standard deviation (s) of the sample's sleep duration have been calculated. The mean sleep duration is $\bar{x}$ = 7.132086 hours, and the standard deviation is s = 0.7956567 hours.

```
> # Sample size
> n <- length(na.omit(data$`Sleep Duration`))
> n
[1] 374
> # Sample mean
> xbar <- mean(data$`Sleep Duration`, na.rm = TRUE)
> xbar
[1] 7.132086
> # Sample standard deviation
> s <- sd(data$`Sleep Duration`, na.rm = TRUE)
> s
[1] 0.7956567
```

Figure 3.1.1: Calculation of $\bar{x}$ and s using RStudio

Using RStudio, the test statistic (t_0), the critical value ($t_{0.05,374}$), and the probability value (p-value) have been calculated as follows: t_0 = -21.095, $t_{0.05,374}$ = -1.648938, and p-value = 6.9148×10^{-66}.

```
> # Hypothesized population mean
> mu <- 8
> mu
[1] 8
> # Calculate t-statistic
> t <- (xbar - mu) / (s / sqrt(n))
> t
[1] -21.09536
> # Calculate p-value for left-tailed test
> pval <- pt(t, df = n - 1, lower.tail = TRUE)
> pval
[1] 6.914882e-66
> alpha = 0.05
> t.alpha = qt(alpha, df = n-1)
> t.alpha
[1] -1.648949
```

Figure 3.1.2: Calculation of t_0, $t_{0.05,374}$ and p-value using RStudio

Values	
alpha	0.05
mu	8
n	374L
pval	6.91488188894087e-66
s	0.795656730889819
t	-21.0953615618475
t.alpha	-1.64894902570761
xbar	7.13208556149733

Figure 3.1.3: All values that calculated using RStudio

Conclusion

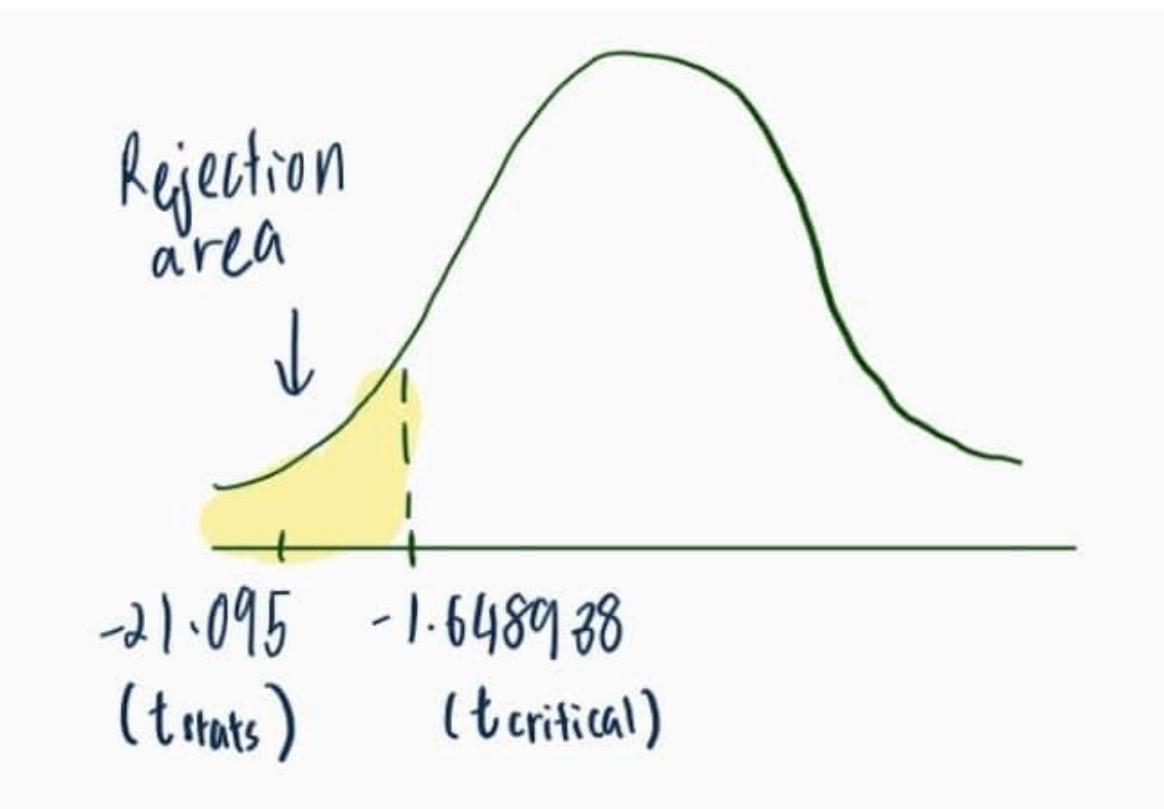


Figure 3.1.4: Graph for one sample hypothesis

Since t_0 = -21.095 < $t_{0.05,374}$ = -1.648938, we reject null hypothesis H_0 : μ=8 at the 0.05 significance level. There is sufficient evidence to support the claim that the mean sleep duration of the population is less than the recommended 8 hours per day.

3.2 Correlation Analysis

In this part of the analysis, the variables chosen are:

- Sleep Duration
- Stress Level

Our aim is to examine the linear relationship between Sleep Duration and Stress Level to understand whether the number of hours a person sleeps is related to their stress level at a scale of 1-10. **Pearson's Product-Moment Correlation Coefficient** is used because Sleep Duration is ratio data and Stress Level is interval data, both variables are numeric. The analysis is conducted at a 95% confidence level, using a sample size of 374 individuals with complete data for both variables. The outcome of this analysis includes the sample correlation coefficient, r, the test statistic and scattered plots as visual representation. The scattered plots along with further explanation of the results will be presented in the following sections.

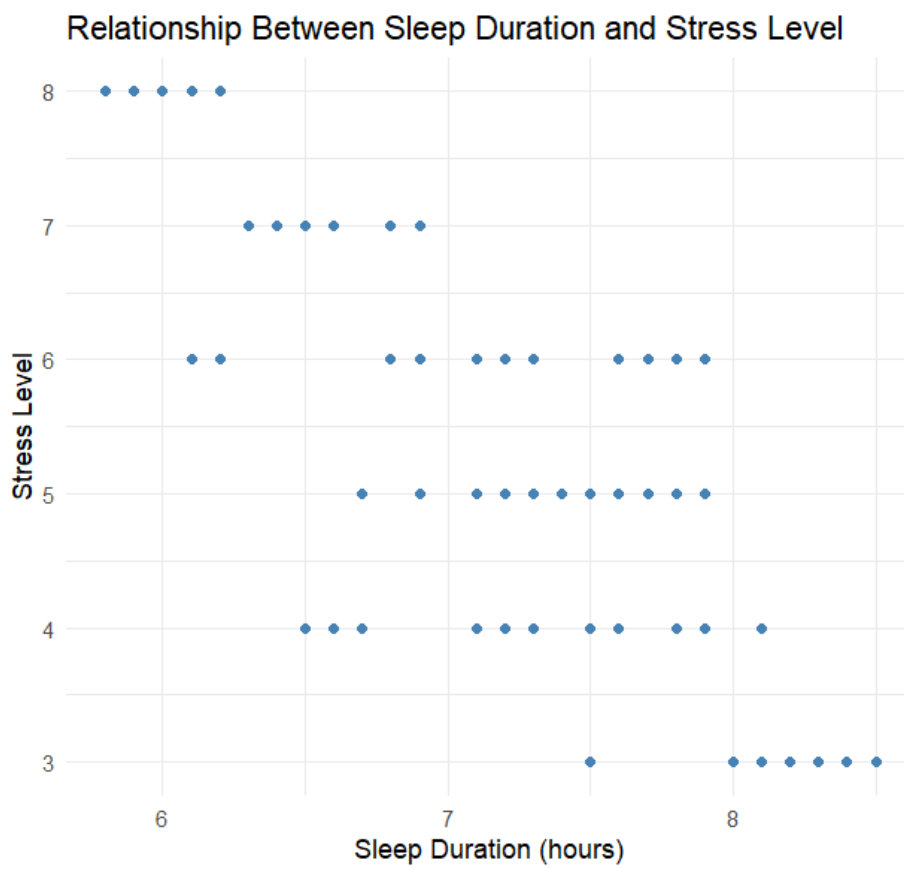


Figure 3.2.1: Scatter plot showing the relationship between Sleep Duration and Stress Level

Figure 3.2.1 shows a visual representation of the relationship between Sleep Duration and Stress Level. However, the scattered plot appears to contain fewer than the expected 374 data points.

This is because some of the points overlap when multiple individuals share similar values for both variables. To address this, another scatter plot was generated using jittering function, to slightly offset and adjust the position of overlapped points. This improves visibility of point patterns and density.

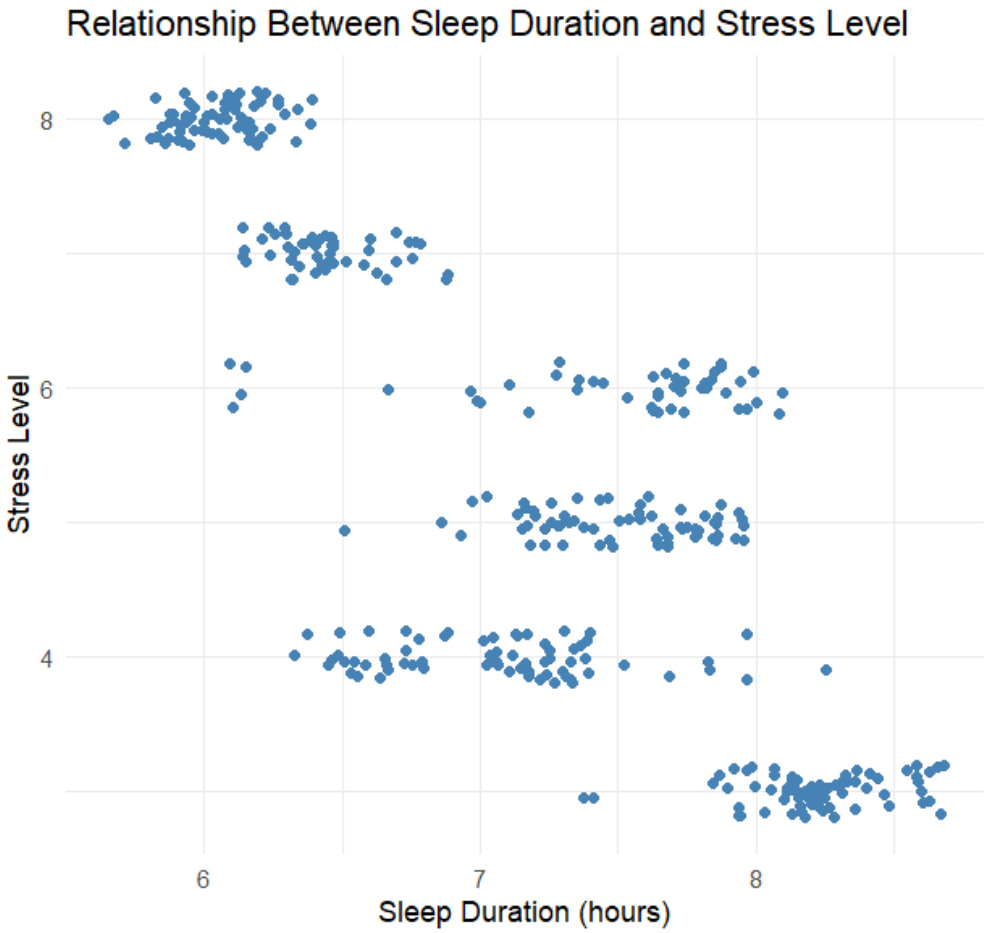


Figure 3.2.2 Jittered scatter plot showing the relationship between Sleep Duration and Stress Level

Figure 3.2.2 shows another visual representation of the relationship between Sleep Duration and Stress Level, this time with reduced overlapping through jittering. The x-axis represents the sleep duration in hours, while the y-axis shows the stress level of individuals. The plot reveals a strong negative correlation between the two variables, indicating that individuals who sleep longer tend to have lower stress levels.

1. **Calculate sample correlation coefficient using Pearson's Method**

Sample correlation coefficient, r = $\dfrac{\Sigma xy - (\Sigma x \Sigma y)/n}{\sqrt{[(\Sigma x^2) - (\Sigma x)^2/n][(\Sigma y^2) - (\Sigma y)^2/n]}}$

Where:
r = Sample correlation coefficient
n = Sample size
x = Value of the independent variable
y = Value of the dependent variable

Calculating the sample correlation coefficient using RStudio:

```
> correlation <-cor(Sleep_health_and_lifestyle_dataset$`Sleep Duration`,
+                   Sleep_health_and_lifestyle_dataset$`Stress Level`, method =
"pearson")
> print(paste("Sample correlation coefficient = ", correlation))
[1] "Sample correlation coefficient =  -0.811023027894043"
```

RStudio is used to calculate the sample correlation coefficient as the sample size of 374 is too large for manual calculation. The result obtained:

Sample correlation coefficient, $r = -0.811023$

From the result shown, the sample correlation coefficient is -0.811023, indicating a strong negative linear association between Sleep Duration and Stress Level. This means that as sleep duration increases, stress level decreases. Individuals who get more sleep generally experience lower levels of stress.

2. **Significance Test for Correlation**

Hypothesis statement:

H_0: $\rho = 0$ $(no\ linear\ correlation)$

H_1: $\rho \neq 0$ $(linear\ correlation\ exists)$

Calculate test statistic using test statistic formula:

t = $\dfrac{r}{\sqrt{\dfrac{1-r^2}{n-2}}}$

Calculating the test statistic using RStudio:

```
> #calculate test statistic
> n<-374
> r<-correlation
> teststat<-r/sqrt((1-r^2)/(n-2))
> print(paste0("Test Statistic = ", teststat))
[1] "Test Statistic = -26.738573017065"
```

By using RStudio to calculate the test statistic, the result obtained:

Test statistic, $t = -26.7386$

The degree of freedom needs to be calculated as well in order to obtain the t critical values from the t-table.

$$\begin{aligned} Degree\ of\ freedom,\ df &= n - 2 \\ &= 374 - 2 \\ &= 372 \end{aligned}$$

Since this is a two-tailed test, significance level, α needs to be divided by 2.

$$\begin{aligned} Significance\ level &= \alpha/2 \\ &= 0.05/2 \\ &= 0.025 \end{aligned}$$

From the t-table, obtain the lower tail critical value and higher tail critical value.

$$-t_{0.025,\,372} = -1.96$$

$$t_{0.025,\,372} = 1.96$$

Calculating the critical value using RStudio:

```
> alpha<-0.05
> df<-n-2
> critval<-qt(1-alpha/2, df)
> print(paste("Critical value at 95% confidence level: ", critval))
[1] "Critical value at 95% confidence level:  1.96636150360549"
```

By using RStudio to calculate the critical value, the result obtained:

$$Critical\ value = \pm 1.9664$$

Hence, if test statistic > 1.9664 or test statistic < -1.9664, reject null hypothesis, H_0. Otherwise, fail to reject null hypothesis, H_0.

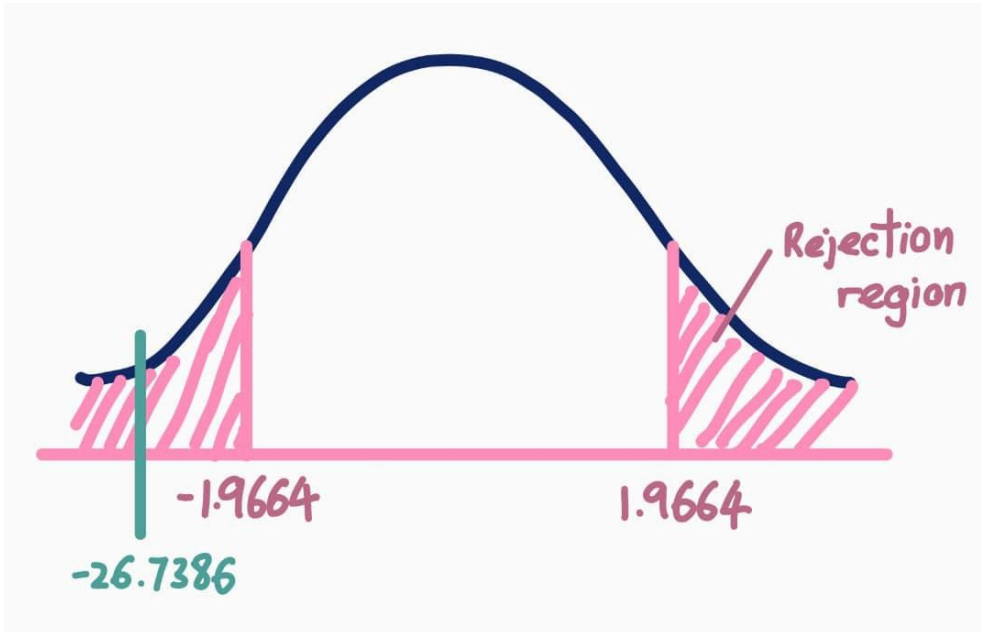


Figure 3.2.3 Bell curve showing the position of the test statistic in the rejection region at 95% confidence level

Since the test statistic, t = - 26.7386 < - $t_{0.025,\,372}$ = - 1.9664, we will **reject** the null hypothesis. In conclusion, there is sufficient evidence to conclude that there is a linear relationship between Sleep Duration and Stress Level at the 95% confidence level.

Performing correlation significance test using RStudio:

```
> cor.test(Sleep_health_and_lifestyle_dataset$`Sleep Duration`,
+          Sleep_health_and_lifestyle_dataset$`Stress Level`,
+          method = "pearson")

        Pearson's product-moment correlation

data:  Sleep_health_and_lifestyle_dataset$`Sleep Duration` and Sleep_health_and_lifestyle_
dataset$`Stress Level`
t = -26.739, df = 372, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8430912 -0.7732074
sample estimates:
      cor
-0.811023
```

3.3 Regression Analysis

In this part of the analysis, we aim to investigate the **linear relationship** between the independent variable, **Physical Activity Level (x)** and the dependent variable, **Quality of Sleep (y)**. Our goal is to determine if an individual's reported physical activity level is a significant predictor of their perceived quality of sleep.

Simple linear regression will be employed for this analysis. This statistical technique is appropriate as Quality of Sleep is interval data and Physical Activity Level is ratio data. Both of the variables are numeric variables.

The analysis will be conducted at a 95% confidence level while utilizing a sample size of 374 individuals with complete data for both variables. The anticipated outcomes of this regression analysis include the regression equation, the coefficient of determination and the test statistic for the overall model's significance.

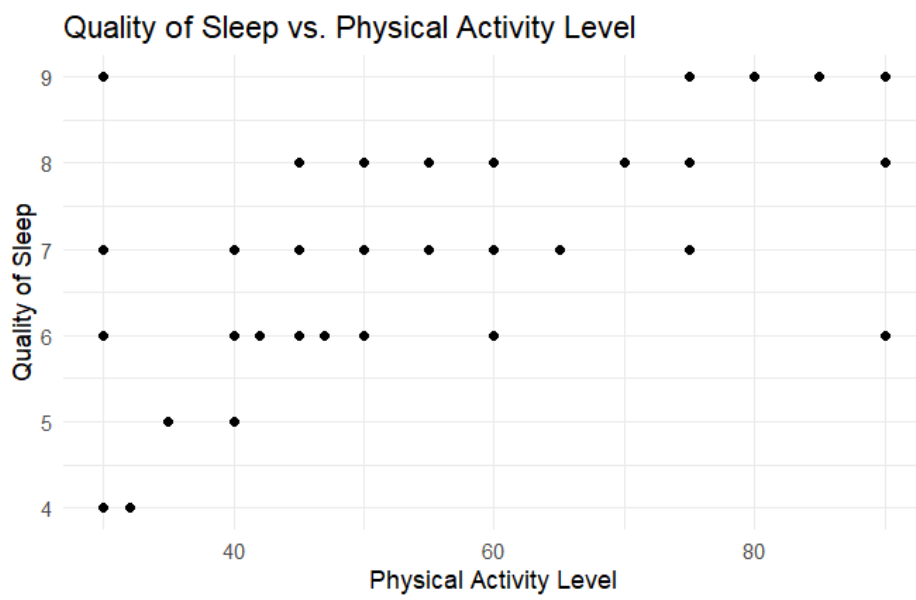


Figure 3.2.4: Scatter plot showing the relationship between Physical Activity Level and Quality of Sleep

Figure 3.2.4 shows a visual representation of the relationship between Physical Activity Level and Quality of Sleep. However, the scattered plot contains less than the expected 374 data points. This is because some of the points overlap when multiple individuals share similar values for both variables. To address this, another scatter plot was generated using jittering function to slightly offset and adjust the position of overlapped points. This improves visibility of point patterns and density.

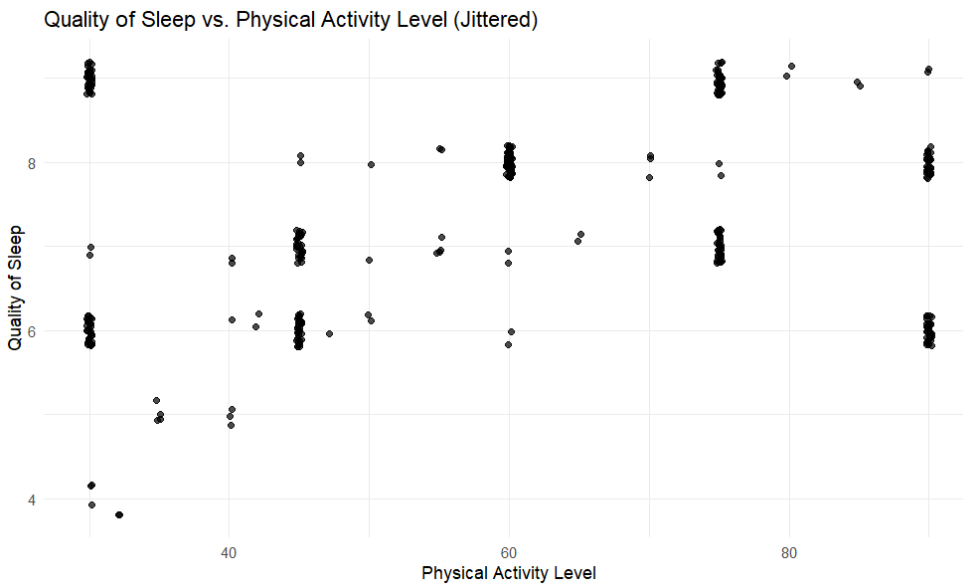
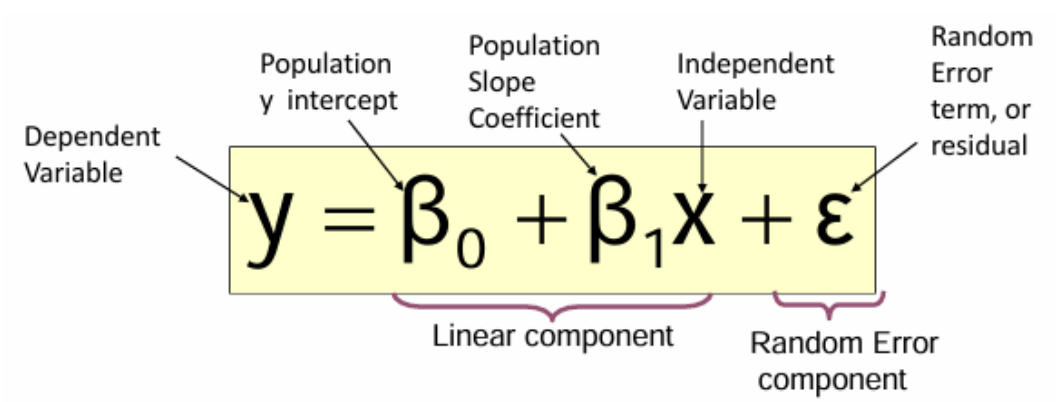


Figure 3.2.5 Jittered scatter plot showing the relationship between Physical Activity Level and Quality of Sleep

Figure 3.2.5 presents a visual representation of the relationship between **Physical Activity Level** (on the x-axis) and **Quality of Sleep** (on the y-axis). This particular scatter plot utilizes jittering to reduce overlapping data points, thereby offering a clearer view of individual observations and revealing patterns that might otherwise be obscured.

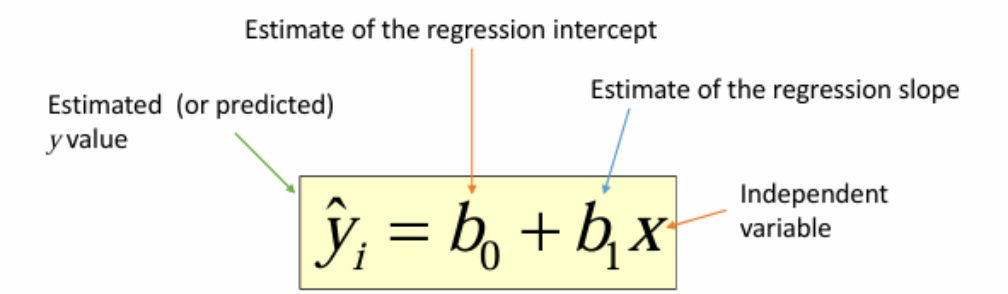
The plot reveals a **positive correlation** between the two variables, indicating that individuals with higher physical activity levels generally tend to report a higher quality of sleep. The spread and density of points suggest that while this positive trend is evident, there can still be variability in sleep quality even among those with similar activity levels.



Our assumptions:

- Error value (ε) are statistically independent and is normally distributed for any value of x.
- The probability of the errors is normal and has constant variance.
- The underlying relationship between x variable and the y variable is linear.

1. Estimated Regression Model:



From the equation above, b_0 is the estimated average value of y (Quality of Sleep) when the value of x (Physical Activity Level) is zero (also known as the y-intercept). Whereas b_1 is the estimated change in the average value of y (Quality of Sleep) due to a one-unit change in x (Physical Activity Level).

- Find the least squares criterion:
  From the above formula, we can find b0 and b1 using formula:

$$b_1 = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}} \quad \text{and} \quad b_0 = \bar{y} - b_1 \bar{x}$$

By using RStudio, we get b_1 = 0.011 and b_0 = 6.657

```
1                (Intercept) 6.656982
2 `Physical Activity Level` 0.011084
```

Substitute the values of b0 and bi into the regression model equation:

ŷi = 6.657 + 0.011x

From the equation, we can interpret the data of the intercept coefficient b_0 and the slope coefficient b_1. In the data, no patient has a Physical Activity Level of 0. Therefore, b_0=6.657 indicates that for patients within the observed range of Physical Activity Level, 6.657 is the estimated portion of Quality of Sleep not explained by Physical Activity Level. Whereas

b_1=0.011 indicates that the average value of Quality of Sleep increases by 0.011 on average, for each additional one-unit increase in Physical Activity Level.

- Explained and Unexplained Variation:

$$SST = SSE + SSR$$

Total sum of Squares	Sum of Squares Error	Sum of Squares Regression
$SST = \sum (y - \bar{y})^2$	$SSE = \sum (y - \hat{y})^2$	$SSR = \sum (\hat{y} - \bar{y})^2$

where:

$\bar{y}$ = Average value of the dependent variable

y = Observed values of the dependent variable

$\hat{y}$ = Estimated value of y for the given x value

Using RStudio, we get:

SSE = 514.51394

SSR = 19.8845

SSE = 514.51394

```
                          Statistic     Value
1      SST (Total Sum of Squares) 534.39840
2      SSE (Error Sum of Squares) 514.51394
3 SSR (Regression Sum of Squares)  19.88445
```

● Find Coefficient of Determination, R^2, using the formula

$$R^2 = \frac{SSR}{SST} \quad \text{where} \quad 0 \le R^2 \le 1$$

By using RStudio, we get:

Coefficient of Determination (R^2): 0.03720904

```
Coefficient of Determination (R^2): 0.03720904
```

Hence, we can interpret it as **3.72%** of the variation in Quality of Sleep is explained by variation in Physical Activity Level.

● Find Standard Deviation of Regression Slope using the formula:

$$s_{b_1} = \frac{s_\varepsilon}{\sqrt{\sum (x - \bar{x})^2}} = \frac{s_\varepsilon}{\sqrt{\sum x^2 - \frac{(\sum x)^2}{n}}}$$

By using RStudio, we get the Standard Deviation of the Regression Slope:

$S_{b1} = 0.0029$

```
Standard Deviation of Regression Slope (Standard Error of b1): 0.002923255
```

2. Inference about the Slope: t-Test

- Hypothesis Statement:
  H0: $\beta 1 = 0$ (no linear relationship)
  H1: $\beta 1 \neq 0$ (linear relationship does exist)

- Find critical value, using α = 0.05, df = n-2 = 372 Since this is two-tailed test, two critical values gotten from RStudio are:

  $-t_{0.025,\,372} = -1.96$

  $t_{0.025,\,372} = 1.96$

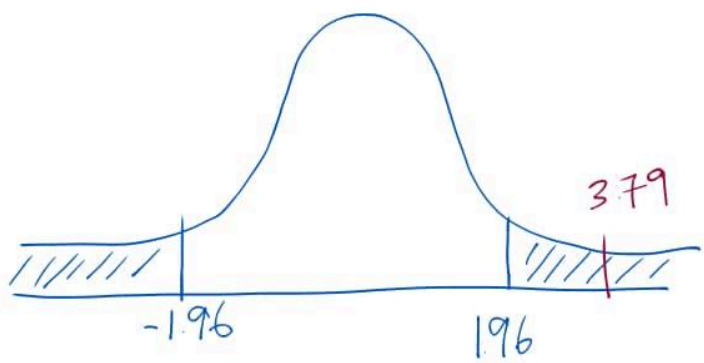
  Hence, we reject H0 if test statistics <-1.96 or test statistics > 1.96.

- Calculate test statistic using formula:

$$t = \frac{b_1 - \beta_1}{s_{b_1}}$$

By using RStudio, we get the test statistic t0: 3.791663

```
Test Statistic (t0): 3.791663
```



Since the test statistic t_0=3.791663 is greater than $t_{0.025,372}$=1.966, we reject the null hypothesis. There is sufficient evidence to prove that Physical Activity Level is linearly related to Quality of Sleep at α=0.05.

∴ Linear Regression Model : $\hat{y}i = 6.657 + 0.011x$

```
Call:
lm(formula = `Quality of Sleep` ~ `Physical Activity Level`,
    data = df)


Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)               6.656982   0.183351  36.307  < 2e-16 ***
`Physical Activity Level` 0.011084   0.002923   3.792 0.000175 ***
```

From the image above, we can see the values of intersection coefficient (intercept) and slope coefficient (x) which is Physical Activity Level.

```
> summary(model)

Call:
lm(formula = `Quality of Sleep` ~ `Physical Activity Level`,
    data = df)

Residuals:
    Min      1Q  Median      3Q     Max
-3.0117 -0.9895 -0.1558  0.6780  2.0105

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)               6.656982   0.183351  36.307  < 2e-16 ***
`Physical Activity Level` 0.011084   0.002923   3.792 0.000175 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.176 on 372 degrees of freedom
Multiple R-squared:  0.03721,	Adjusted R-squared:  0.03462
F-statistic: 14.38 on 1 and 372 DF,  p-value: 0.0001745
```

From this summary of linear regression model, we can obtain values such as

b0 = 6.657
b1 = 0.011
sb1 = 0.0029
sε = 1.176
df = 372
R^2 = 0.0372
p-value = 0.0001745

- Finally, we can add the linear regression model into the scatter plot:

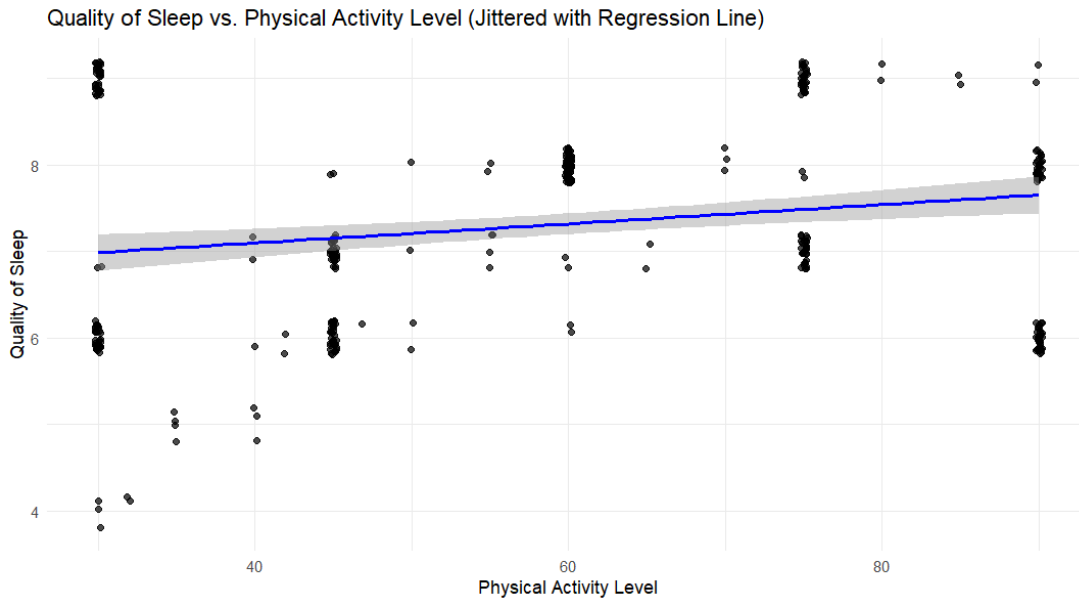


Figure 3.2.6 Linear Regression Model showing the relationship between Physical Activity Level and Quality of Sleep

Figure 3.2.6 presents a visual representation of the relationship between Physical Activity Level (on the x-axis) and Quality of Sleep (on the y-axis), with the added linear regression model. This particular scatter plot utilizes jittering to reduce overlapping data points, thereby offering a clearer view of individual observations and revealing patterns that might otherwise be obscured.

The plot reveals a positive correlation between the two variables, indicating that individuals with higher physical activity levels generally tend to report a higher quality of sleep. The spread and density of points suggest that while this positive trend is evident, there can still be variability in sleep quality even among those with similar activity levels. The superimposed linear regression line further illustrates this overall positive trend, providing a visual summary of the estimated average change in Quality of Sleep for a given Physical Activity Level.

3.4 Chi-Square Test of Independence

In this analysis, we examine the relationship between Gender and Type of Sleep Disorder to determine whether they are independent of each other. Using a two-way contingency table and applying the Chi-Square Test of Independence at the 95% confidence level, we assess whether any statistically significant association exists between the two variables.

```
> contingency_table <- table(sleep_data$Gender, sleep_data$`Sleep Disorder`)
> print(contingency_table)

        Insomnia None Sleep Apnea
  Female       36   82          67
  Male         41  137          11
```

Observed Frequencies for variables type of sleeping disorder and gender

Step 1: State the hypothesis statement

H_0: Both of the variables which is Gender and Type of Sleep Disorder are independent

H_1: Both of the variables which is Gender and Type of Sleep Disorder are dependent

Step 2: Find the critical value

```
> # Calculate degrees of freedom
> df <- (nrow(contingency_table) - 1) * (ncol(contingency_table) - 1)
> print(paste("Degrees of Freedom:", df))
[1] "Degrees of Freedom: 2"
> # Calculate critical value for alpha = 0.05
> alpha <- 0.05
> critical_value <- qchisq(1 - alpha, df)
> print(paste("Critical Value (α = 0.05):", round(critical_value, 4)))
[1] "Critical Value (α = 0.05): 5.9915"
```

Finding degree of freedom and critical value of x^2 using RStudio

Critical value, x^2 = 5.991

Degree of Freedom = 2

Step 3: Calculate the expected counts

```
> # Check expected counts (should be >5 for valid chi-square test)
> print(chi_test$expected)

        Insomnia     None Sleep Apnea
  Female 38.08824 108.3289    38.58289
  Male   38.91176 110.6711    39.41711
```

Table 3.4.1: Table for observation and expectation count

	Type of Sleep Disorder						**Total**
	Insomnia		**None**		**Sleep Apnea**		
	Obs.	**Exp.**	**Obs.**	**Exp.**	**Obs.**	**Exp.**	
Female	36	38.088	82	108.33	67	38.583	185
Male	41	38.912	137	110.67	11	39.417	189
Total	77	77	219	219	78	78	374

Step 4: Calculate the test statistic value

- **Calculate manually:**

Table 3.4.2: Table for test statistic value that calculate manually

Cell, ij	**Observed Count,** O_{ij}	**Expected Count,** e_{ij}	$\frac{(O_{ij} - e_{ij})^2}{e_{ij}}$
1,1	36	$\frac{77\ x\ 185}{374} = 38.088$	$\frac{(36-38.088)^2}{38.088} = 0.115$
1,2	82	$\frac{219\ x\ 185}{374} = 108.33$	$\frac{(82-108.33)^2}{108.33} = 6.400$
1,3	67	$\frac{78\ x\ 185}{374} = 38.583$	$\frac{(67-38.583)^2}{38.583} = 20.930$
2,1	41	$\frac{77\ x\ 189}{374} = 38.912$	$\frac{(41-38.912)^2}{38.912} = 0.1120$
2,2	137	$\frac{219\ x\ 189}{374} = 110.67$	$\frac{(137-110.67)^2}{110.67} = 6.264$
2,3	11	$\frac{78\ x\ 189}{374} = 39.417$	$\frac{(11-39.417)^2}{39.417} = 20.487$
		x^2	54.308

- **Calculate using RStudio:**

```
> print(chi_test)

        Pearson's Chi-squared test

data:  contingency_table
X-squared = 54.306, df = 2, p-value = 1.613e-12
```

When calculate test statistic using RStudio, we get test statistic, x^2 = 54.306 with p-value 1.613×10^{-12}.

Step 5: State the decision

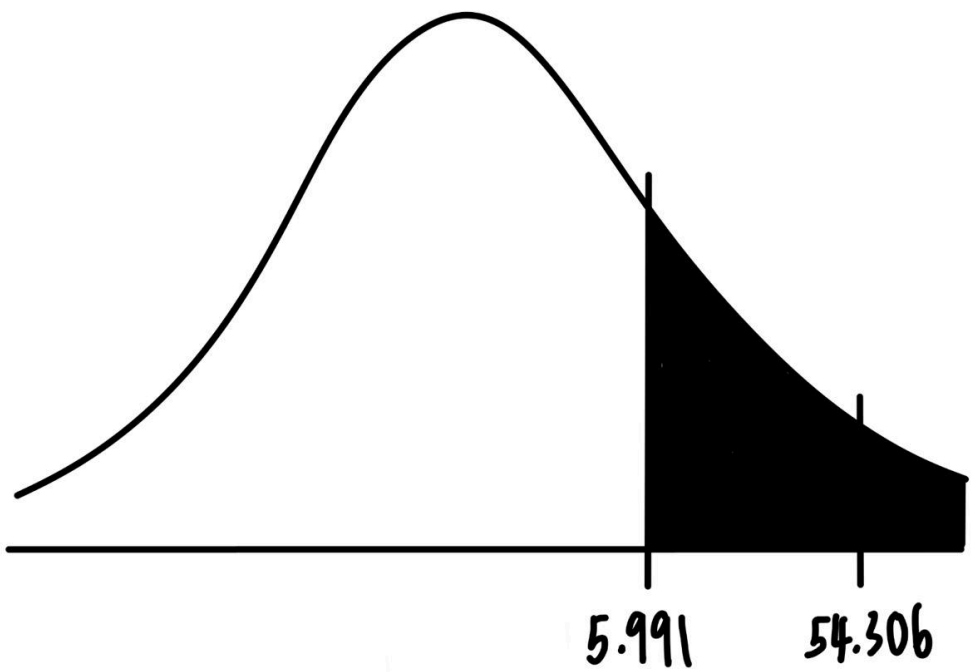


Figure 3.4.1: Graph for Chi-Square Test of Independence

Since the test statistic value (= 54.306) > critical value (= 5.991), it does fall within the critical region. Thus, we reject the null hypothesis, H_0. Therefore, there is sufficient evidence to conclude that there is a relationship between the variable type of sleeping disorder and gender, at α = 0.05.

4.0 Conclusion & Discussion

In the one-sample hypothesis test, we investigated whether the mean sleep duration of the population is equal to the recommended 8 hours per day, as suggested by scientific guidelines. Based on the analysis, the results indicate that the average sleep duration of the sample is significantly less than 8 hours. Therefore, we reject the null hypothesis at the 0.05 significance level. This suggests that, on average, people sleep less than the recommended 8 hours per day. This finding may reflect common sleep deprivation patterns in modern society, which could have implications for health and well-being.

In the correlation test, we examined whether there is a significant linear relationship between Sleep Duration and Stress Level. The analysis showed a strong negative correlation, with a Pearson correlation coefficient of -0.8110. The analysis also gave a test statistic of t = -26.739, which falls far into the rejection region when compared to the critical value of -1.9664 at the lower tail. This means we can confidently reject the null hypothesis and conclude that people who sleep more tend to experience lower stress levels. These findings reflect how important good sleep habits are for managing stress and support the idea that getting enough rest can have a positive impact on mental well-being.

In the Regression Test, we examined whether there is a linear relationship between independent variable, Physical Activity Level and the dependent variable, Quality of Sleep in our dataset of 374 individuals. Simple linear regression is employed for this analysis. The analysis successfully established a statistically significant positive linear relationship between an individual's Physical Activity Level and their Quality of Sleep (t-statistic = 3.972, p-value < 0.05). This indicates that as physical activity level increases, there is a tendency for Quality of Sleep to also increase. The estimated regression model, $\hat{y}i = 6.657 + 0.011x$, quantifies this relationship. The slope coefficient (b_1=0.011) suggests that for every one-unit increase in Physical Activity Level, the estimated average Quality of Sleep increases by 0.011 units. The intercept (b_0=6.657) represents the estimated Quality of Sleep when Physical Activity Level is zero, though its direct interpretation might be limited as no patients in the dataset had a Physical Activity Level of zero. However, the coefficient of determination (R2=0.0372 or 3.72%) indicates that only a very small proportion of the variation in Quality of Sleep can be explained by the variation in Physical Activity Level.

In the Chi-Square Test of Independence, we examined whether there is a significant association between gender and type of sleep disorder in our dataset of 374 individuals. The analysis revealed a highly significant relationship between these two variables. With the test statistic substantially exceeding the critical value of 5.991 at $\alpha = 0.05$, we confidently reject the null hypothesis of independence. The results indicate that gender and sleep disorder type are not independent variables in this population. The extremely small p-value (1.613×10^{-12}) also

provides overwhelming evidence against the null hypothesis, suggesting that the observed differences in sleep disorder patterns between males and females are statistically significant.

The overall findings from our statistical analyses provide strong evidence of significant relationships between sleep-related variables and factors such as stress, physical activity, and gender. On average, individuals sleep less than the recommended 8 hours, which is associated with higher stress levels. Increased physical activity is linked to better sleep quality, although it explains only a small portion of the variation. Additionally, gender is significantly associated with the type of sleep disorder experienced. These insights highlight the importance of promoting healthy sleep habits, physical activity, and targeted interventions to address gender-specific sleep health concerns.

This is a really beneficial and meaningful project to all of us as it provides an opportunity for us to dive deeper into RStudio and R programming, which is an effective tool to analyse both small and big data. This allows us to get prepared to deal with real-life situations where we often need to analyse large amounts of different data. Lastly, we express our appreciation to our lecturer, Dr. Shamini A/P Raja Kumaran, for providing help and guidance along the way.

5.0 Appendix

The dataset used for this analysis was obtained from Kaggle:

Health and Sleep Relation 2024 Dataset
Available at: https://www.kaggle.com/datasets/orvile/health-and-sleep-relation-2024/data

Video Presentation:

https://youtu.be/Z9yKQNTRMMo